

Protected Crop Production

Science

May, 2026

Protected Crop Production (A08621)

Short Title: Protected Crop Production
Faculty Science and Computing
Department Science
Cluster Horticulture
Author GMCMAHON
Credits 5

Level: Intermediate

Description of Module / Aims

The student will be introduced to a range of protected structures and their environmental control. The module will enable the student to draw up management and production schedules for a range of crops in greenhouses using best industry practice.

Programmes

HORT -0048	BSc in Horticulture (WD_SHORT_D)	3 / 6 / E
HORT -0048	BSc in Horticulture (WD_SHORT_DX)	3 / 6 / E

Indicative Content

- Structure, design and site selection of protective structures
- Environmental management and control within protective structures
- Water and nutrient management of protected crops
- Crop production techniques for heated and cold greenhouse crops
- Production and management of a selection of heated and cold greenhouse crops
- Plant protection of protected crops
- Glass House technology and sensor technology, precision technology.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Determine appropriate protective structures for specific site requirements.
2. Propose appropriate environmental control, irrigation and feeding systems.
3. Develop production and management schedules for protected crops.
4. Integrate good environmental practices into recommendations for heated and cold crops.
5. Demonstrate safely a range of skills involved in protected crop production.
6. Investigate contemporary developments in glass house and sensor technology.

Learning and Teaching Methods

- Lectures
- Practical demonstration
- Practice

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,4,6
Continuous Assessment	50%	
<i>Practical</i>	25%	5
<i>Project</i>	25%	3

Assessment Criteria

- <40%:** Very limited knowledge of protected plant production methods and cannot demonstrate understanding of production methods or protected structures. Unable to complete practical skills or exercises.
- 40%-49%:** Demonstrates a limited knowledge of protected plant production systems and processes. Limited technical ability but demonstrates ability to learn and understand new skills.
- 50%-59%:** Demonstrates satisfactory general knowledge of the main areas of protected plant production. Competently carries out practical tasks and skills. Reports and projects adequately structured and referenced.
- 60%-69%:** Demonstrates sound knowledge of protected production principles and systems. Demonstrates ability to logically interpret and analyse technical information. Projects, reports and exams coherent and soundly structured.
- 70%-100%:** Demonstrates authoritative handling of complex material and a highly developed and extensive understanding of the subject. Demonstrates imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/professional tasks.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Supplementary Material

- Beytes, C. *_Ball Redbook Volume 1: Greenhouses and Equipment_*. 18th ed. Illinois: Ball Publishing, 2011.
- Boodley, J.W. and S.E. Newman. *_The Commercial Greenhouse_*. 3rd ed. New York: Delmar Cengage Learning, 2009.
- MacNeidhe, F. *_Out of Season Strawberry Production under Polythene_*. Dublin, Ireland. : Teagasc, 1999.